COVID VISUAL ANALYTICS PORTAL

An Interactive Exploration Platform

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Executive Summary

- ✓ COVID Visual Analytics Portal: an **interactive web platform** to explore the pandemic’s impact on professional life & public health.
- ✓ Combines **survey data**, **state stats**, and **global metrics** into a **unified dashboard** at one place.
- ✓ Features **interactive maps, dynamic charts, predictive insights**.
- ✓ Empowers **researchers & policymakers** with **clear, actionable trends**.
- ✓ Built for **scalability & future expansion** — a reusable model for future health or economic crises.
- ✓ Built with a robust **Python backend** + **ReactJS & D3.js frontend** for seamless analysis.

Project Overview & Goals

- ✓ Unprecedented impact on work & health
- ✓ Sudden remote work shift, stress, productivity changes
- ✓ Goals:
  - Analyze professional life impacts
  - Monitor pandemic stats
  - Combine individual & public data
  - Support policymakers & researchers
  - Enable interactive, user-friendly exploration
  - Ensure scalability & reuse for future crises
  - Promote multi-level storytelling (personal → regional → global)

Scope

- ✓ Survey Data: Stress, productivity, WFH
- ✓ State Stats: Indian COVID-19 cases
- ✓ Global Comparison: India vs world
- ✓ Interactive Dashboard: Seamless data exploration
- ✓ Multi-Dimensional Analysis: Individual, regional, global
- ✓ Dynamic Visualizations: Maps, charts, scatter plots, heatmaps
- ✓ Cross-Sector Insights: Trends across different work sectors
- ✓ Integrated Data Pipeline: Clean data, reliable stories

Technologies Used

- ✓ Backend: Python (pandas, re, seaborn, matplotlib)
- ✓ Frontend: ReactJS, D3.js, HTML, CSS
- ✓ Tools: Git, GitHub, VS Code

Tasks Addressed

- • Analyze stress & WFH trends
- • Visualize productivity
- • State-level COVID-19 stats
- • Global comparisons
- • Interactive storytelling

Data Sources & Preprocessing

- ✓ • cleaned_data.csv: Survey (stress, WFH, productivity)
- ✓ • state_data.csv: Indian states stats
- ✓ • worldometer_data.csv: Global stats
- ✓ • Python scripts: clean_data.py, create_data3.py
- ✓ • Cleaning: regex, fill missing, normalize

Application Architecture

- • Backend: Data cleaning & prep, CSVs
- • Frontend: ReactJS UI, D3.js charts, responsive maps & graphs
- • Clear pipeline: raw data → interactive storytelling

Overall Application Story

- • Connects global, regional, individual impacts
- • India’s global standing, state trends, personal impact
- • Maps, charts, EDA tell clear story

Tabs & Features

- • Survey Data Tab: Stress, productivity, working hours
- • State-Level Tab: COVID in states
- • Global Tab: India vs world
- • Charts: Pie, bar, scatter, heatmap

Global Employment Impact

- • Disrupted jobs worldwide
- • Metrics: working hours, gender, labor dependency
- • Findings: Immediate losses, regional/gender gaps
- • Predictive model: Random Forest (R^2=0.78)

Contributions

- • Kartik: Backend scripts
- • Lakshyta: Survey, EDA visuals
- • Manvi: State/global features
- • Ravi: ReactJS, D3.js
- • Pranav: Frontend-backend link
- • Chetna: API, debugging
- • Bhaskar: Maps, tooltips
- • Priyanshu: Cleaning, EDA plots
- • Deepak: Styling, presentation

Challenges & Limitations

- • Data inconsistency
- • Self-report quality
- • Frontend performance
- • Deployment constraints
- • Clear UI vs info-rich visuals

Future Work

- • ML for forecasting
- • Real-time updates
- • Broader surveys
- • Advanced visuals
- • Scalable deployment

Conclusion

- • Unified platform: Public health + professional impact
- • Combines cleaned data & dynamic visuals
- • Supports policymakers, researchers, public
- • Extensible for future

Code & Demo Links

- • GitHub: github.com/Manvi-56/covid-visual-analytics
- • Live Dashboard: cs661-covid-data-analysis.netlify.app

Thank You

- Questions & Discussions